

1. ALGORITHM

Main Algorithm:

1. Start
2. Display menu: 1. Strong Number, 2. Armstrong Number, 3. Harshad Number, 4. Duck Number
3. Input choice
4. Switch(choice)
 - Case 1: Call checkStrong()
 - Case 2: Call checkArmstrong()
 - Case 3: Call checkHarshad()
 - Case 4: Call checkDuck()
 - Default: Print invalid choice
5. Stop

Functions Algorithm:

- **checkStrong()**: Take input number. Extract digits using loop. Find factorial of each digit using a simple for loop. Add factorials. If sum equals number, it is Strong number.
- **checkArmstrong()**: Take input number. Count digits using a loop. Calculate power of each digit using `pow()` from `math.h`. Add them up. If sum equals number, it is Armstrong number.
- **checkHarshad()**: Take input number. Find sum of its digits using a loop. If number is divisible by sum of digits, it is Harshad number.
- **checkDuck()**: Take input number. Extract digits using a loop. If any digit is 0, set flag = 1. If flag is 1, it is Duck number.

2. TEST CASES TABLE

Test Case	Operation	Input	Expected Output	Result
1	Strong Number	145	145 is a Strong Number	Pass
2	Strong Number	123	123 is NOT a Strong Number	Pass
3	Armstrong Number	153	153 is an Armstrong Number	Pass
4	Armstrong Number	370	370 is an Armstrong Number	Pass
5	Harshad Number	18	18 is a Harshad Number	Pass
6	Harshad Number	19	19 is NOT a Harshad Number	Pass
7	Duck Number	3210	3210 is a Duck Number	Pass
8	Duck Number	432	432 is NOT a Duck Number	Pass

3. C PROGRAM CODE

```
#include <stdio.h>
#include <math.h>

void checkStrong();
void checkArmstrong();
void checkHarshad();
void checkDuck();

int main() {
    int choice;

    printf("
--- MENU ---
");
    printf("1. Check Strong Number
");
    printf("2. Check Armstrong Number
");
    printf("3. Check Harshad Number
");
    printf("4. Check Duck Number
");
    printf("Enter choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            checkStrong();
            break;
        case 2:
            checkArmstrong();
            break;
        case 3:
            checkHarshad();
            break;
        case 4:
            checkDuck();
            break;
        default:
            printf("Invalid choice
");
    }

    return 0;
}

void checkStrong() {
```

```

int num, temp, rem, sum = 0, fact, i;

printf("Enter a number: ");
scanf("%d", &num);

temp = num;
while(temp > 0) {
    rem = temp % 10;

    fact = 1;
    for(i = 1; i <= rem; i++) {
        fact = fact * i;
    }

    sum = sum + fact;
    temp = temp / 10;
}

if(sum == num) {
    printf("%d is a Strong Number", num);
} else {
    printf("%d is NOT a Strong Number", num);
}
}

void checkArmstrong() {
    int num, temp, rem, count = 0, sum = 0;

    printf("Enter a number: ");
    scanf("%d", &num);

    temp = num;
    while(temp > 0) {
        count++;
        temp = temp / 10;
    }

    temp = num;
    while(temp > 0) {
        rem = temp % 10;
        sum = sum + (int)pow(rem, count);
        temp = temp / 10;
    }

    if(sum == num) {
        printf("%d is an Armstrong Number", num);
    } else {

```

```
        printf("%d is NOT an Armstrong Number  
", num);  
    }  
}
```

```
void checkHarshad() {  
    int num, temp, rem, sum = 0;  
  
    printf("Enter a number: ");  
    scanf("%d", &num);  
  
    temp = num;  
    while(temp > 0) {  
        rem = temp % 10;  
        sum = sum + rem;  
        temp = temp / 10;  
    }  
  
    if(num % sum == 0) {  
        printf("%d is a Harshad Number  
", num);  
    } else {  
        printf("%d is NOT a Harshad Number  
", num);  
    }  
}
```

```
void checkDuck() {  
    int num, temp, rem, flag = 0;  
  
    printf("Enter a number: ");  
    scanf("%d", &num);  
  
    temp = num;  
    while(temp > 0) {  
        rem = temp % 10;  
        if(rem == 0) {  
            flag = 1;  
            break;  
        }  
        temp = temp / 10;  
    }  
  
    if(flag == 1) {  
        printf("%d is a Duck Number  
", num);  
    } else {  
        printf("%d is NOT a Duck Number  
", num);  
    }  
}
```

```
}  
}
```

4. SAMPLE OUTPUT

Output 1:

--- MENU ---

1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number

Enter choice: 1

Enter a number: 145

145 is a Strong Number

Output 2:

--- MENU ---

1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number

Enter choice: 3

Enter a number: 18

18 is a Harshad Number